

Jack Bulson

Gameplay Programmer

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Personal Profile

Programmer with 4 years experience in the games industry. I have a passion for gameplay and UI programming. Proficient in C++, with experience shipping multiplayer and single-player titles. Highly motivated and looking for new experiences within the games industry.

Employment

Gameplay Programmer (C++ / Unreal Engine)

Red Rover Interactive
April 2025 ~ June 2026

- Worked on **Enginefall**, a multiplayer PVP survival crafting/extraction game.
- Responsible for building system **design** and **UX** improvements including refactoring the building system to support **prefabs** (collections of building pieces) and making placing objects easier.
- Implemented a Tech Tree System and UI. This supported both individual and shared progression. The UI was built to be performant to support **large numbers** of individual tech-tree nodes.
- Created a world marker system which pulls data about points of interest within the game from multiple **dynamic** and **static** sources to provide data for an in-game compass and map to use.
- Reacted quickly and efficiently to **player feedback** and **bug reports** from weekly public playtests.

Programmer (C++ / Unreal Engine)

Sumo Newcastle
Jan 2024 ~ March 2025

- Worked on **Critter Cafe**, a single-player critter collecting, cafe management cozy game.
- Implemented **multiple** gameplay mechanics to a **high-standard** in C++, these include: character's tools, level ingredients(blocks/redirectors/switches), cafe open for trade + minigame, **UI**.
- Created these mechanics from a **specification**, and worked collaboratively with designers & level designers to **iterate** on them and develop **tooling** to solve workflow issues.
- Responsible for reworking the customisation mode to improve the **game feel** and **UX**.
- Experience gained with **fixing bugs** and **debugging** across a complex game with data used across a variety of places, such as save data or global game data.

Graduate Programmer (C++ / Unreal Engine)

Sumo Newcastle
May 2022 ~ Jan 2024

- Worked on **DeathSprint 66**, a **multiplayer**, high-speed on-foot racing game.
- Prototyped gameplay systems and mechanics within a **small feature team**, iterating on these to find the fun and moving on once we were satisfied with the outcome.
- Worked on a **shelved** project where I took the lead on implementing outlines for interactable objects, improving the **player experience** and **accessibility**.

Projects

Rendering Project (C++ / Vulkan)

2025

- Ported the rendering portion of my game engine to gain more experience with C++ (without Unreal)
- Created my own format that I could serialize meshes, materials, scenes and images into. This is so that they were optimised for the data formats within my project and easier to load at run-time.
- Implemented basic SSAO

Game Engine (Rust / Vulkan)

2022 ~ 2023

- Consisted of a renderer, simple asset management, collision detection and other utilities
- Used **data-driven design** to create a performant rendering pipeline
- Implemented lighting, shadows, post-processing, instanced rendering, compute shaders
- Created a renderlist that used virtual resources to **abstract** away render passes

Technical Skills & Qualifications

- 4 years industry experience using C++/Unreal Engine/Perforce
- MComp Computer Science (Games Engineering) Newcastle University, UK (1st)